

MCMC sampling over a density — random-walk Metropolis

propose $\mathbf{x}' \sim \mathcal{N}(\mathbf{x}^{(t)}, \sigma^2 \mathbf{I})$, accept w.p. $\alpha = \min(1, p(\mathbf{x}')/p(\mathbf{x}^{(t)}))$

target $p(\mathbf{x}) \propto \exp(-E(\mathbf{x}))$ — a bimodal mixture; the chain crosses the low-density valley

